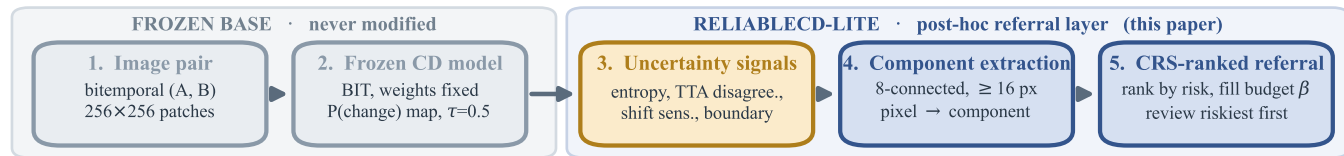


ReliableCD-Lite

training-free · post-hoc · no model retraining

A post-hoc layer that ranks predicted change components by error risk and spends a fixed analyst review budget on the riskiest regions.



The same post-hoc layer, organized as a three-layer hierarchy — the protocol is the spine:

(i) Component-level protocol

- Review unit = one predicted change component.
- Rank components by score; greedily fill area budget β (1% / 3% / 5% / 10% / 20%).
- Analyst inspects the highest-risk regions first.
- Component- vs. pixel-level referral at the same reviewed area.

PRIMARY CONTRIBUTION

reviewed within budget β



↑ score orders the components

(ii) CRS scoring — the ranking instrument

- CRS-3 (fast, one forward pass): $z(U_{\text{ent}}) + z(U_{\text{bnd}}) - z(m)$
- CRS-4 (full): CRS-3 + $z(U_{\text{tta}}) + z(U_{\text{shift}})$
- Equal weights; z-scores fit once on source-domain validation; no target-domain labels.

↑ signals feed the score

(iii) Uncertainty signals — post-hoc diagnostics



SCOPE Only predicted components are scored — completely missed changes (false negatives) lie outside the ranked, observable set.